WIND-AWARE MARITIME SURVEILLANCE

Nusantara SolarWing

Wind-Aware Reserve Planning for Natuna Maritime Patrols



CONCEPT RENDERING / NOT A PROTOTYPE

EXECUTIVE SUMMARY / ABSTRACT

Indonesia's remote maritime sectors need repeatable surveillance, yet a small electric UAV can lose its return margin when clouds reduce solar input and headwinds raise return demand. Nusantara SolarWing proposes a solar-assisted fixed-wing aircraft and a wind-aware reserve planner for a conceptual patrol from Ranai, Natuna. The planner maximizes new search-line distance as a coverage proxy, subject to a protected battery reserve. At each waypoint it combines battery state, conservative solar input, and high-side return-energy estimates to command patrol, shorten, or return. A five-leg creeping-line pattern defines a 12 km by 8 km planning box with 2 km lane spacing, beginning 20 km from the Ranai reference point. A first-principles airframe model estimates 147.5 W level-flight draw and an 894 Wh nominal 6S12P battery. NASA POWER hourly irradiance near Natuna for September 2025 yields 29 of 30 days passing an eight-hour, no-wind energy screen at a 20% reserve. That count is not a route safety result. For a 20 km return, an assumed 5 m/s headwind raises modeled energy from 60.7 to 96.4 Wh. With the same pack, battery-only nominal endurance to reserve is 4.03 hours, versus up to eight modeled daylight hours with solar assistance. Bench, weather-replay, and supervised flight tests will verify power, forecast calibration, and coverage before operational use.

01 / MODEL	02 / MODEL	03 / MODEL
29 / 30	**+59%**	**4.03 h**
historical no-wind days pass	20 km return energy at 5 m/s headwind	battery-only limit to 20% reserve

CLAIM BOUNDARY

These are scenario calculations. No aircraft prototype, patrol permission, vessel detector, or flight-test result is claimed.

01 / PHYSICS + EVIDENCE

Size the energy budget

A fixed-wing solar platform trades speed and payload for endurance. The proposed 13.8 kg, 5.2 m span airframe has 1.82 m^2 wing area and 1.65 m^2 active photovoltaic area. It carries a conceptual camera payload for human maritime review, with 14.5 W allocated to avionics and payload. No sensor coverage is assumed [2, 3].

AIRFRAME AND ELECTRICAL MODEL

Cruise / density	13.5 m/s / 1.18 kg/m^3	Level flight, no wind
Wing / span	1.82 m^2 / 5.20 m	Aspect ratio 14.86
Drag assumptions	Cd0 0.022 / Oswald e 0.82	Not wind-tunnel verified
Prop / motor / ESC	0.82 / 0.88 / 0.95	Efficiency assumptions
PV / MPPT	1.65 m^2 / 24% / 96.5%	Horizontal-GHI proxy
Battery	6S12P NCR18650GA	894.24 Wh nominal

DERIVATION

CL = 2mg/(rho V^2S) = 0.692; CD = Cd0 + CL^2/(pi AR e) = 0.0345; L/D = 20.05. Propulsion electrical power = drag × V/(0.82 × 0.88 × 0.95) = 133.0 W. With payload and avionics, level-flight load = **147.5 W**.

HISTORICAL SOLAR SCREEN

NASA POWER hourly all-sky surface irradiance at 4.0°N, 108.0°E covers September 2025 in WIB after shifting the UTC series +7 hours [1]. Median daily GHI = 5.24 kWh/m^2. Model 08:00–16:00 at 147.5 W, 90% starting SoC, 20% reserve, and 95% charge/discharge efficiency: **29/30 historical days pass**. The weakest day reaches 15.9% SoC. Median theoretical PV generation is 2.00 kWh/day, while 24-hour continuous demand is 3.54 kWh/day.

MODEL BOUNDARY

The 29/30 count is a no-wind energy screen, not route feasibility or a reliability estimate. It omits climb, turns, shading, battery aging and return uncertainty. A 6S12P pack also needs structural, thermal and mass verification.

02 / RESERVE-AWARE PLANNER

A decision at every waypoint

Objective: maximize $\Sigma L_i x_i$, the length of newly completed search lines; x_i is a binary leg choice. This proxies coverage until camera swath is measured. The hard constraint protects the return route and 20% reserve. Ties favor lower energy. Conservative forecast bounds are design targets, not calibrated predictions [6]. All margin terms are battery-equivalent Wh.

DECISION FLOW

01 / OBSERVE

Battery energy, solar input, wind and position

02 / BOUND

Use low-side solar and high-side return energy

03 / COMPUTE

M = Ebat + Esolar,low - Ereturn,high - Ereserve - Ebuffer

PATROL

M ≥ full next-leg energy

SHORTEN

0 < M < full next-leg energy

RETURN

M ≤ 0 or safety trigger

EXAMPLE THRESHOLD AT 20 KM RANGE

At 20 km, assume 5 m/s return headwind, zero future-solar credit and a 12 km next leg. At 147.5 W with 95% battery discharge efficiency: return debit = 101.5 Wh, next-leg debit = 38.3 Wh, reserve = 178.8 Wh and 10% buffer = 89.4 Wh. Thus **return at ≤369.7 Wh (41.3% SoC)**, shorten at 369.7–408.1 Wh, and patrol at ≥408.1 Wh (45.6% SoC). Recompute from actual position; a short leg still needs a safe exit.

FAIL-SAFE GUARD

If measured SoC falls below reserve, wind exceeds the safe return bound, or telemetry/forecast fails, command immediate return or pre-planned alternate landing. The operator retains authority. The controller may authorize a leg only when its predicted return margin stays positive throughout that leg.

03 / MARITIME MISSION + IMPACT

Start with a bounded sector

Ranai Airport's published reference point is 03°55′12″N, 108°23′08″E [4]. We use it as a geometric anchor only. A conceptual offshore search box begins 20 km from that point. The actual launch site, airspace, coast clearance and sector coordinates require operator and authority review.

CREEPING-LINE SEARCH PATTERN



Five 12 km parallel legs plus four 2 km lane changes form a 12 × 8 km planning box. Straight segments total 68 km. With at least 20 km ingress and 20 km egress, **108 km is a route lower bound**; the exit may be farther from base. Turning arcs, launch and recovery add distance. The 96 km² box is an assigned footprint, not verified camera coverage; sensor swath and overlap must be measured.

20 KM RETURN LEG / HEADWIND SENSITIVITY

HEADWIND	GROUND SPEED	RETURN TIME	ENERGY	VS CALM
0 m/s	13.5 m/s	24.7 min	**60.7 Wh**	baseline
5 m/s	8.5 m/s	39.2 min	**96.4 Wh**	+59%
8 m/s	5.5 m/s	60.6 min	**149.0 Wh**	+145%

Model: time = 20 km/(13.5 m/s minus headwind); energy = 147.5 W × time. It assumes constant airspeed, load and wind aligned with the return. A headwind at or above 13.5 m/s removes positive ground speed in this simplified model and triggers diversion planning.

QUANTIFIED BENEFIT / MATCHED-PACK COMPARATOR

At 90% launch SoC and a 20% reserve, the same 894 Wh battery alone supports **4.03 h** of modeled 147.5 W level flight after 95% discharge efficiency. Solar assistance extends the no-wind screen to 8 h on 29/30 historical days, an upper scenario of **+3.97 h (+98%)**. An eight-hour sortie draws 1.18 kWh at the aircraft. A battery-only equivalent would require about 1.31 kWh grid input at 95% discharge and charge efficiency. Monetary and CO_2 savings require a measured fuel or local-grid comparator; none is claimed.

20 KM SECTOR CHECK / IDEAL ROUTE LOWER BOUND

For 20 km calm outbound + 68 km calm sweep + 20 km return into 5 m/s headwind, with zero solar credit, battery debit is 382.5 Wh. From 90% launch SoC, the pack ends at 47.2%, 17.2 points above reserve plus buffer. The exit may be farther than 20 km; turns, climb and other losses still require testing.

04 / FEASIBILITY + VERIFICATION

Test the decisions, not the slogan

Energy-aware solar navigation has prior art [6]. The proposed contribution is a measurable Natuna maritime route policy that combines conservative solar input, wind-sensitive return energy and explicit shorten/return thresholds. No claim of algorithmic superiority is made before a common-trace comparison.

VALIDATION GATES

01 / BENCH	Measure propulsion W versus airspeed, payload draw, PV output under shading, MPPT and battery usable Wh.
02 / REPLAY	Use weather traces outside September 2025 to calibrate low-solar and high-wind bounds. Compare fixed-time and reserve-aware policies on coverage, minimum SoC and forced returns.
03 / FLIGHT	Fly a subscale demonstrator under pilot oversight. Check predicted versus observed energy, sensor swath, turn losses and link-failure response.

OPERATIONAL AND IMPACT LIMITS

One month at one NASA grid point cannot establish annual mission availability. Surface horizontal GHI differs from PV irradiance aloft. Constant propulsion power ignores gust response and maneuvers. The search-box footprint is not detection coverage. Fuel, cost and emissions benefits versus boats or combustion UAVs cannot be quantified without a matched mission and measured comparator. AIS absence alone does not imply illegal activity [5]. Airspace approval remains separate.

REFERENCES

[1] NASA POWER. Hourly API; Sep 2025 ALLSKY_SFC_SW_DWN at 4.0°N, 108.0°E.

[2] Panasonic Energy. NCR18650GA Li-ion cell specifications, 3.60 V and 3.45 Ah typical.

[3] Noth, A. (2008). Design of Solar Powered Airplanes for Continuous Flight. ETH Zürich.

[4] Indonesian Directorate General of Civil Aviation. Ranai Airport reference point.

[5] International Maritime Organization. AIS carriage and operation guidance.

[6] Energy-Aware 3D Navigation of a Solar-Powered UAV in an Urban Environment. ICCAE 2024.

[7] ISAST 2026 Aerospace Innovation Challenge. Track and document rules.